

Mohammad Aman Ullah Khan

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SUMMARY

Research-focused Computer Science graduate specializing in Large Language Models (LLMs), Natural Language Processing (NLP), Machine Learning, and AI systems. Experienced in transformer architectures, multimodal learning, explainable AI, and clinical AI applications. Published researcher with work in Elsevier, IEEE, Springer, and EAI on healthcare imaging, speech translation, and intelligent systems. Research interests include privacy-preserving, multilingual, and multimodal AI.

EDUCATION

BRAC University

BSc in Computer Science and Engineering, CGPA: 3.29/4.0

Dhaka, Bangladesh

July 2019 - Aug 2024

Thesis Title: Bangla Dialect Speech to Standard English Speech Translation for Rural Telemedicine

Supervisors: Dr. Md. Golam Rabiul Alam and Md. Tanzim Reza

RESEARCH EXPERIENCE

- Research Assistant** Karnataka, India
School of Computer Engineering, Manipal Institute of Technology (MIT), MAHE Jul 2025 – Present
 - Contributed to the AI-Centre of Excellence in Health (AI-COE) at Manipal Academy of Higher Education, a nationally funded AI-in-healthcare research initiative supported by the Ministry of Education, Government of India.
 - Conducted research on AI-driven healthcare analytics, medical imaging, and clinical decision support systems.
- Research Assistant** Andhra Pradesh, India
School of Computer Science and Engineering (SCOPE), VIT-AP University Jun 2024 – Jul 2025
 - Published three IEEE papers on deep learning applications in medical imaging, precision agriculture and cybersecurity.
 - Designed hybrid CNN and attention-based architectures, optimized transfer learning strategies, and conducted Hyperband and Bayesian hyperparameter optimization for image classification tasks.
 - Mentored junior researchers in model development, experimentation, performance evaluation, and scientific writing.
- Machine Learning Research Intern** Andhra Pradesh, India
School of Computer Science and Engineering (SCOPE), VIT-AP University Dec 2023 – Jun 2024
 - Implemented deep learning pipelines using transfer learning, data augmentation, and hyperparameter optimization.
 - Evaluated model performance through experimentation and image preprocessing for computer vision applications.
- Undergraduate Researcher** Dhaka, Bangladesh
Computer Science and Engineering, School of Data and Sciences, BRAC University Jan 2023 – Mar 2024
 - Led the development of a Bangla dialect-to-English speech translation framework for rural telemedicine using NLP, LLMs, and neural machine translation for real-time medical communication.
 - Collected and augmented 50+ hours of Bengali speech data for dialect-aware healthcare translation applications.

PUBLICATIONS

- Kiran, A., Ramesh, J. V. N., Rahat, I. S., **Khan, M. A. U.**, Hossain, A., & Uddin, R. (2024). Advancing breast ultrasound diagnostics through hybrid deep learning models. *Computers in Biology and Medicine*, 180, 108962. DOI: 10.1016/j.compbimed.2024.108962
- Zerine, I., Islam, M. M., **Khan, M. A. U.**, Chy, M. A. R., Saimon, A. S. M., Manik, M. M. T. G., & Wata, C. (2026). Explainable churn prediction in telecom with tabular ML: Five model benchmark and SHAP analysis. *Discover Artificial Intelligence*, 6, 263. DOI: 10.1007/s44163-026-00983-0
- Islam, M. M., Zerine, I., Hossain, M. Z., **Khan, M. A. U.**, Shekhor, K. C., Islam, M., & Wata, C. (2026). Combining semantic CNN and HOG texture with LightGBM for explainable tuberculosis detection. *Discover Artificial Intelligence*, 6, 409. DOI: 10.1007/s44163-026-01109-2
- Ghosh, H., Rahat, I. S., Ravindra, J. V. R., J, B., **Khan, M. A. U.**, & Somasekar, J. (2024). Convolutional neural networks in malaria diagnosis: A study on cell image classification. *EAI Endorsed Transactions on Pervasive Health and Technology*, 10. DOI: 10.4108/eetpht.10.5551
- Porna, S. B., Kabir, M. F., Rana, M. I. C., Sajol, M. S. I., Roy, T., **Khan, M. A. U.**, Adnan, M. A., & Bhavani, G. D. (2024). Hybrid convolutional neural networks for enhanced detection of mango leaf diseases. *2024 IEEE 6th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, 547–552. DOI: 10.1109/ICCCMLA63077.2024.10871711
- Sharma, R., Banik, S., Hossain, M. Z., Kabir, M. F., Thapa, S., **Khan, M. A. U.**, Hossain, A., Uddin, R., & Bhavani, G. D. (2024). A novel classification scheme for early Alzheimer's disease (AD) severity diagnosis using deep features of a hybrid cascade attention architecture. *2024 International Conference on Augmented Reality, Intelligent Systems, and Industrial Automation (ARIIA)*, 1–6. DOI: 10.1109/ARIIA63345.2024.11051777
- Talukder, M. S. H., Nur, A. H., Zaman, S., Noor, M. R., **Khan, M. A. U.**, & Amir, F. (2024). Optimizing diabetes prediction accuracy: A comprehensive approach with advanced preprocessing and diverse machine learning classifiers. *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 1–6.

DOI: 10.1109/ICAEEE62219.2024.10561634

- Porna, S. B., Hossain, M. Z., Raj, M. G., Uddin, R., Elahi, M. A., Basit, A., **Khan, M. A. U.**, & Bhavani, G. D. (2024). Advanced malware classification using convolutional neural networks and image transformation techniques. *2024 2nd International Conference on Advancements and Key Challenges in Green Energy and Computing (AKGEC)*, 1–5.
DOI: 10.1109/AKGEC62572.2024.10868701

Accepted / Under Review

- **Khan, M. A. U.**, Gope, N., & Alam, G. R. (2025). Bangla Dialect Speech to Standard English Speech Translation System for Rural Telemedicine Application in Bangladesh [Manuscript submitted]. *IEEE Transactions on Neural Networks and Learning Systems*. DOI: 10.5281/zenodo.19172115
- **Khan, M. A. U.**, & Rahat, I. S. (2025). Advancing Brain Cancer & Spinal Cord Tumor Detection: Innovations, Challenges, & Future Directions [Accepted for publication]. *Expert Systems with Applications (ESWA)*.
DOI: 10.5281/zenodo.19170989
- **Khan, M. A. U.**, Rahat, I. S., & Kiran, A. (2025). Fusion of Deep Learning and Transfer Learning in BrainNet-SVM for Superior Brain Tumor Detection [Accepted for publication]. *Machine Learning*.
DOI: 10.5281/zenodo.19171484
- **Khan, M. A. U.**, Khasim, S., & Rahat, I. S. (2026). Privacy-Preserving Federated Brain Tumor Segmentation using nnU-Net and Swin-UNETR across Multiple Institutions [Accepted for presentation]. *Innovations in Machine, Engineering and Digital (IMED) Conference*.
DOI: 10.2139/ssrn.5922122

PROFESSIONAL SERVICE

- Journal Peer Reviewer, *Biomedical Signal Processing and Control* (Elsevier) – Reviewed 3 manuscripts (2024–Present)
- Conference Reviewer, 4th IEEE International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE 2025), 2025

TEACHING EXPERIENCE

- **Teaching Assistant, Backend Engineering** Dhaka, Bangladesh
• *Tecognize Solutions Limited* Dec 2021 – Dec 2023
 - Instructed the “Backend Engineering with Python” course, mentoring 200+ students in Python, Java, SQL, HTML.
 - Coordinated Python backend projects and cloud deployment activities; developed technical blogs and presentations.

OTHER PROJECTS

- **Drone Weather Station (IoT, Sensors, Embedded Systems)**: Engineered an autonomous drone with Arduino sensors to collect real-time atmospheric data using sensor calibration and embedded systems programming. Tech: C++, Arduino, APM 2.8, Mission Planner, 3DR Tower, GPS, Ardupilot & BLDC Motors (May '21)
- **Next Gen Mood Detection via Augmented Google-FER (Deep Learning & Computer Vision)**: Developed a CNN model to classify six emotions from eye images using custom FER-2013 dataset and image preprocessing. Tech: Python, TensorFlow, Keras, OpenCV, CNNs & Data Augmentation (Oct '20)

HONORS AND AWARDS

- Vice Chancellor’s List, BRAC University (Summer 2020)
- Government Talent Pool Scholarship, PSC, JSC, and SSC Examinations (2011, 2014, 2016)

CERTIFICATIONS AND TRAININGS

- **Applied Machine Learning**
Bangladesh-Korea Information Access Center, Department of CSE, BUET, Duration: 8 weeks, Year: 2025
- **Mathematics for Machine Learning Specialization**
Imperial College London on Coursera, Duration: 4 months, Year: 2024
- **2024 ClinGen Somatic Cancer and VICC Virtual Molecular Tumor Board On Demand**
American College of Medical Genetics and Genomics, Duration: 8 months, Year: 2024
- **Python 3 Programming Specialization**
University of Michigan on Coursera, Duration: 4 months, Year: 2024

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, R, SQL

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, OpenCV, Scikit-learn

Tools & Platforms: Git, LaTeX, MATLAB, MySQL, Arduino/Embedded Systems